

Geometry

Choose the appropriate Answer

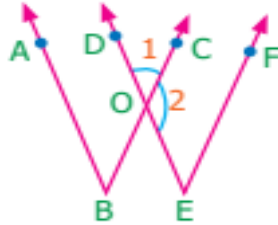
1. A straight path that goes on endlessly in two directions.
(a) Ray (b) Line (c) Line Segment (d) None
Ans: (b) Line
2. A line with two end points.
(a) Ray (b) Line (c) Line Segment (d) None
Ans: (c) Line Segment
3. A _____ is a straight path that begins at a point and goes on and extends endlessly in the other direction.
(a) Ray (b) Line (c) Line Segment (d) None
Ans: (a) Ray
4. The lines which intersect at right angles are _____ lines.
(a) Perpendicular (b) Parallel (c) Intersecting (d) None
Ans: (a) Perpendicular
5. The lines which intersect each other at a point are called _____.
(a) Perpendicular (b) Parallel (c) Intersecting (d) None
Ans: (c) Intersecting
6. The lines that never intersect are called _____.
(a) Perpendicular (b) Parallel (c) Intersecting (d) None
Ans: (b) Parallel
7. A straight angle measures
(a) 45° (b) 90° (c) 180° (d) 100°
Ans: (c) 180°
8. An angle with measure 128° is called _____ angle.
(a) a straight (b) an obtuse (c) an acute (d) Right
Ans: (b) an obtuse
9. The corner of the A4 paper has
(a) Acute angle (b) Right angle (c) Straight (d) Obtuse angle
Ans: (b) Right angle
10. If a perpendicular line is bisecting the given line, you would have two
(a) Right angles (b) Obtuse angles (c) Acute angles (d) Reflex angles
Ans: (a) Right angles
11. An angle that measure 0° is called _____.
(a) Right angle (b) Obtuse angle (c) Acute angle (d) Zero angle
Ans: (d) Zero angle

12. Adjacent angles have

- (a) No common interior, no common arm, no common vertex
- (b) One common vertex, one common arm, common interior
- (c) One common arm, one common vertex, no common interior
- (d) One common arm, no common vertex, no common interior

Ans: (c) One common arm, one common vertex, no common interior

13. In the given figure the angles $\angle 1$ and $\angle 2$ are



- (a) Opposite
- (b) Adjacent
- (c) Linear pair
- (d) Supplementary

Ans: (c) Linear pair

14. Vertically opposite angles are

- (a) Not equal in measure
- (b) Complementary
- (c) Supplementary
- (d) Equal in measure

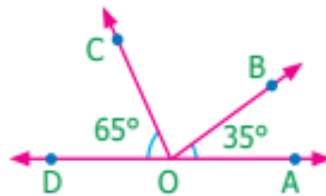
Ans: (d) Equal in measure

15. The sum of all angles at a point is

- (a) 360°
- (b) 180°
- (c) 90°
- (d) 0°

Ans: (a) 360°

16. The measure of $\angle BOC$ is



- (a) 90°
- (b) 180°
- (c) 80°
- (d) 100°

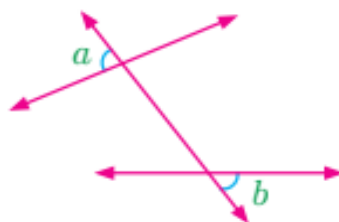
Ans: (c) 80°

17. A line which intersects two or more lines in different points is known as ----- lines.

- (a) Parallel
- (b) Transversal
- (c) Non-parallel
- (d) Intersecting

Ans: (b) Transversal

18. In the given figure, angles a and b are



- (a) Alternate exterior angles
- (b) Corresponding angles
- (c) Alternate interior angles
- (d) Vertically opposite angles

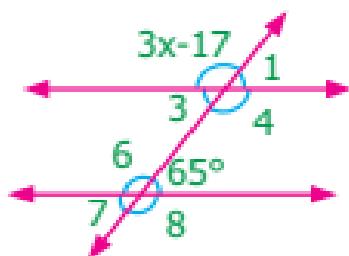
Ans: (a) Alternate exterior angles

19. Which of the following statements is ALWAYS TRUE when parallel lines are cut by a transversal

- (a) Corresponding angles are supplementary.
- (b) Alternate interior angles are supplementary.
- (c) Alternate exterior angles are supplementary.
- (d) Interior angles on the same side of the transversal are supplementary.

Ans: (d) Interior angles on the same side of the transversal are supplementary.

20. In the diagram, what is the value of angle x?



- (a) 43°
- (b) 44°
- (c) 132°
- (d) 134°

Ans: (b) 44°

21. When the sum of the measures of two angles is 90° , the angles are called -----

- (a) Supplementary angles
- (b) Complementary angles
- (c) Adjacent angles
- (d) Vertically opposite angles

Ans: (b) Complementary angles

22. The measure of the complement of the angle 30° is

- (a) 30°
- (b) 16°
- (c) 60°
- (d) 160°

Ans: (c) 60°

23. Which one of the following statements is not false?

- (a) If two angles form a linear pair, then each of these angles is of measure 90° .
- (b) Angles forming a linear pair can both be acute angles.
- (c) One of the angles forming a linear pair can be an obtuse angle.
- (d) Bisectors of the adjacent angles form a right angle.

Ans: (c) One of the angles forming a linear pair can be an obtuse angle.

24. The measure of the complement of the angle 46° is

- (a) 90°
- (b) 44°
- (c) 16°
- (d) 136°

Ans: (b) 44°

25. What is the measure of the complement of the angle 80° is

- (a) 10°
- (b) 100°
- (c) 36°
- (d) 20°

Ans: (a) 10°

26. The measure of the angle which is equal to its complement is

- (a) 30°
- (b) 60°
- (c) 45°
- (d) 90°

Ans: (c) 45°

27. Which of the following pairs of angles is not a pair of complementary angles?
(a) 60° , 30° (b) 66° , 34° (c) 0° , 90° (d) 150° , 30°

Ans: (d) 150° , 30°

28. What is the measure of the complement of the angle 90° ?

(a) 90° (b) 0° (c) 180° (d) 46°

Ans: (b) 0°

29. When the sum of the measures of two angles is 180° , the angles are called

(a) Adjacent angles (b) Complementary angles
(c) Vertically opposite angles (d) Supplementary angles

Ans: (d) Supplementary angles

30. The measure of the supplement of the angle 120° is

(a) 30° (b) 45° (c) 60° (d) 90°

Ans: (c) 60°

31. The measure of the supplement of the angle 90° is

(a) 45° (b) 60° (c) 30° (d) 90°

Ans: (d) 90°

32. Which of the following pairs of angles is not a pair of supplementary angles?

(a) 90° , 90° (b) 32° , 58° (c) 0° , 180° (d) 76° , 104°

Ans: (b) 32° , 58°

33. The angles in a linear pair are

(a) Complementary angles (b) Supplementary angles
(c) Not adjacent angles (d) Vertically opposite angles

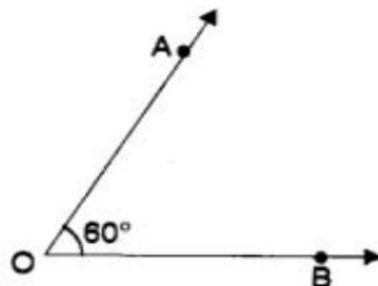
Ans: (b) Supplementary angles

34. Which of the following statements is false?

(a) Two vertically opposite angles can be acute.
(b) Two vertically opposite angles can be obtuse.
(c) Two vertically opposite angles can be right angles.
(d) Two vertically opposite angles may be unequal.

Ans: (d) Two vertically opposite angles may be unequal

35. In the following figure, reflex $\angle AOB$ is equal to

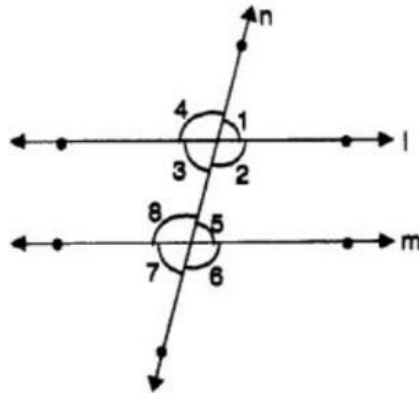


(a) 60° (b) 120° (c) 300° (d) 360°

Ans: (c) 300°

$$\angle AOB = 360^\circ - 60^\circ = 300^\circ$$

36. In the following figure, tell which pair of angles are not corresponding angles?

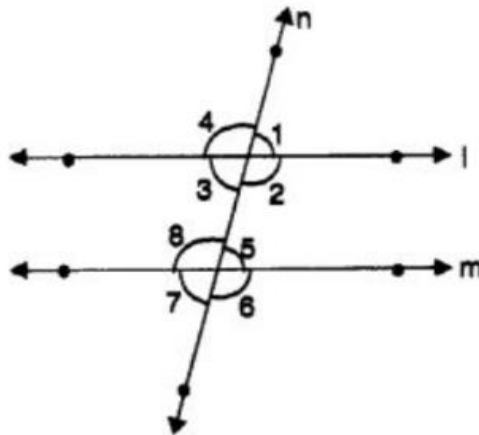


- (a) $\angle 1, \angle 5$ (b) $\angle 2, \angle 6$ (c) $\angle 3, \angle 7$ (d) $\angle 3, \angle 5$

Ans: (d) $\angle 3, \angle 5$

$\angle 3, \angle 5$ are alternate interior angles

37. In the following figure, a transversal cuts two parallel lines l and m respectively and the angles thus formed are marked. If $\angle 1$ is an acute angle, then, which of the following statements is false?



- (a) $\angle 1 + \angle 2 = 180^\circ$ (b) $\angle 2 + \angle 5 = 180^\circ$ (c) $\angle 3 + \angle 8 = 180^\circ$ (d) $\angle 2 + \angle 6 = 180^\circ$

Ans: (d) $\angle 2 + \angle 6 = 180^\circ$

$\angle 2 = \angle 6$, corresponding angles

38. Find the measure of the angle which is double its complementary angle?

- (a) 60° (b) 30° (c) 45° (d) 120°

Ans: (a) 60°

$$x^\circ = 2(90^\circ - x^\circ) \Rightarrow x^\circ = 60^\circ$$

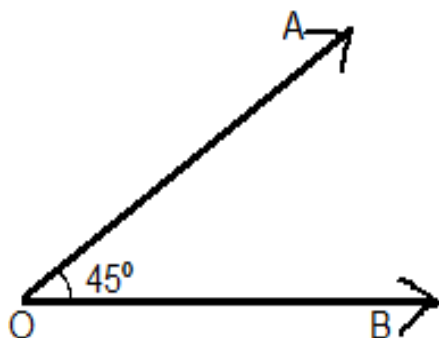
39. Find the measure of the angle which is half of its supplementary angle?

- (a) 60° (b) 120° (c) 90° (d) 45°

Ans: (a) 60°

$$x^\circ = \frac{1}{2}(180^\circ - x^\circ) \Rightarrow x^\circ = 60^\circ$$

40. In the following figure, reflex $\angle AOB$ is equal to



- (a) 60° (b) 120° (c) 315° (d) 360°

Ans: (c) 315°

$$\angle AOB = 360^\circ - 45^\circ = 315^\circ$$

41. A line that intersects two or more lines at distinct points is called ----- lines.

- (a) Parallel (b) Transversal (c) Intersecting (d) None of these

Ans: (b) Transversal

42. When an arm of an angle is extended then the measure of angle:

- (a) Doubles (b) Triples (c) Remains the same (d) None

Ans: (c) Remains the same

An arm will not affect the measure of an angle.

43. Complementary pair of angles:

- (a) $70^\circ + 20^\circ$ (b) $30^\circ + 45^\circ$ (c) $0^\circ + 95^\circ$ (d) None of these

Ans: (a) $70^\circ + 20^\circ$

44. Identify supplementary pair of angles:

- (a) $0^\circ, 90^\circ$ (b) $30^\circ, 150^\circ$ (c) $45^\circ, 145^\circ$ (d) None of these

Ans: (b) $30^\circ, 150^\circ$

45. What do we call an angle whose measurement is exactly equal to 0° ?

- (a) An obtuse angle (b) A straight angle (c) A zero angle (d) A right angle

Ans: (c) A zero angle

46. Which instrument is used to measure or construct angles?

- (a) Compasses (b) Scale (c) Protractor (d) Set

Ans: (c) Protractor

47. In the given figure angles a and b are
(a) Alternate exterior angles (b) Corresponding angles
(c) Alternate interior angles (d) Vertically opposite angles

Ans: (a) Alternate exterior angles

48. A line which intersects two or more lines in different points is known as ---- lines.

(a) Parallel (b) Non-parallel (c) Transversal (d) Intersecting

Ans: (b) Non – parallel

49. Vertically opposite angles are

(a) Not equal in measure (b) Complementary
(c) Supplementary (d) Equal in measure

Ans: (d) Equal in measure

50. Adjective angles have

(a) No common interior, no common arm, no common vertex.
(b) One common vertex, one common arm, common interior
(c) One common arm, one common vertex, no common interior.
(d) One common arm, no common vertex, no common interior.

Ans: (d) One common arm, one common vertex, no common interior.

51. The sum of all angles at a point is

(a) 360° (b) 180° (c) 90° (d) 0°

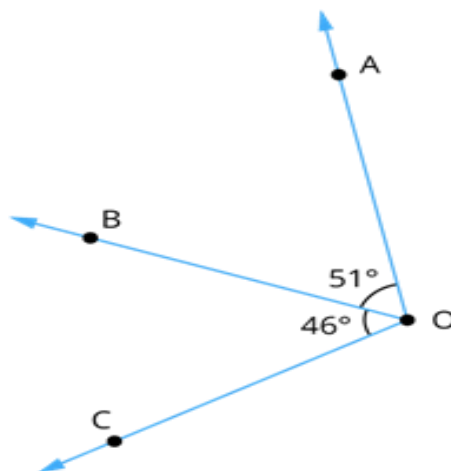
Ans: (a) 360°

52. Which of the following statement is ALWAYS TRUE when parallel lines are cut by a transversal

(a) Corresponding angles supplementary.
(b) Alternate interior angles supplementary.
(c) Alternate exterior angles supplementary.
(d) Interior angles on the same side of the transversal are supplementary.

Ans: (d) Interior angles on the same side of the transversal are supplementary.

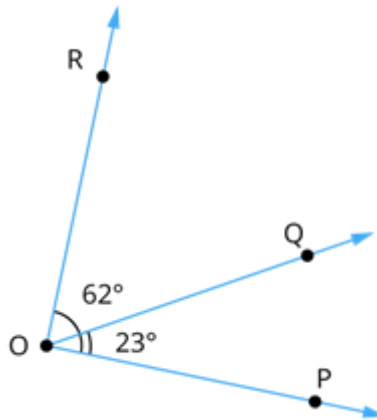
53. In the given figure, find $\angle AOC$.



(a) 97° (b) 99° (c) 98° (d) 96°

Ans: (a) 97°

54. If $\angle POQ = 23^\circ$ and $\angle POR = 62^\circ$ then find $\angle QOR$.



- (a) 85° (b) 28° (c) 39° (d) 67°
Ans: (c) 39°

55. If two lines intersect at a point, then the vertically opposite angles are always

- _____
(a) Equal (b) Unequal (c) Supplementary (d) Complementary
Ans: (a) Equal

56. Two angles forming a linear pair are _____.

- (a) Equal (b) Unequal (c) Supplementary (d) Complementary
Ans: (c) Supplementary

57. A line that intersects two or more lines at distinct points is called

- (a) Parallel (b) Transversal (c) Intersecting (d) None of these
Ans: (b) Transversal

58. If two adjacent angles are supplementary, then they form _____.

- (a) Corresponding angles (b) Vertically opposite angles
(c) A linear pair of angles (d) A ray
Ans: (c) A linear pair of angles

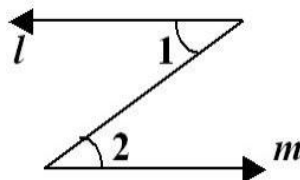
59. If two angles are supplementary then the sum of their measures is -----.

- (a) 90° (b) 180° (c) 360° (d) 45°
Ans: (b) 180°

60. If two angles are complementary, then the sum of their measures is -----.

- (a) 45° (b) 180° (c) 90° (d) 360°
Ans: (c) 90°

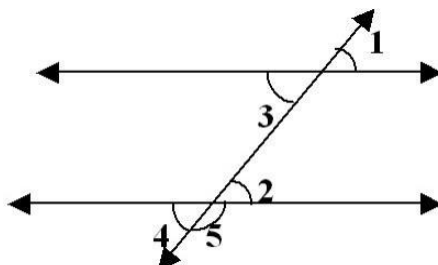
61. If $l \parallel m$, then $\angle 1 = \angle 2$ because they are _____.



- (a) Corresponding angles (b) Vertically opposite angles
(c) Alternate interior angles (d) Supplementary angles

Ans: (c) Alternate interior angles

62. In fig. pair of alternate interior angles are:



- (a) $\angle 1, \angle 3$ (b) $\angle 2, \angle 3$ (c) $\angle 2, \angle 5$ (d) $\angle 1, \angle 2$

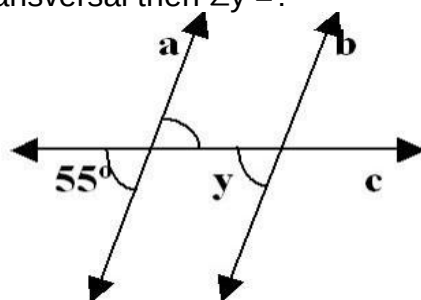
Ans: (b) $\angle 2, \angle 3$

63. If two parallel lines are cut by a transversal, each pair of the corresponding angles are _____ in measure.

- (a) Equal (b) Unequal (c) Supplementary (d) Complementary

Ans: (a) Equal

64. Line $a \parallel b$, c is a transversal then $\angle y = ?$



- (a) 90° (b) 125° (c) 55° (d) 180°

Ans: (c) 55°

65. The difference in the measures of two complementary angles is 12° . Find the measures of the angles.

- (a) 51° and 49° (b) 51° and 39° (c) 60° and 30° (d) 50° and 40°

Ans: (b) 51° and 39°

66. What is the measure of the complement of 45° ?

- (a) 135° (b) 25° (c) 35° (d) 45°

Ans: (d) 45°

67. What is the measure of the complement of 65° ?

- (a) 135° (b) 25° (c) 35° (d) 45°

Ans: (b) 25°

68. What is the measure of the complement of 41° ?

- (a) 139° (b) 49° (c) 35° (d) 45°

Ans: (b) 49°

69. What is the measure of the supplement of 54° ?

- (a) 126° (b) 49° (c) 35° (d) 41°

Ans: (a) 126°

70. Identify which of the following pairs of angles are complementary

- (a) $65^\circ, 115^\circ$ (b) $63^\circ, 27^\circ$ (c) $112^\circ, 68^\circ$ (d) $130^\circ, 50^\circ$

Ans: (b) $63^\circ, 27^\circ$

71. Identify which of the following pairs of angles are supplementary.

- (a) $80^\circ, 10^\circ$ (b) $63^\circ, 27^\circ$ (c) $112^\circ, 68^\circ$ (d) $45^\circ, 45^\circ$

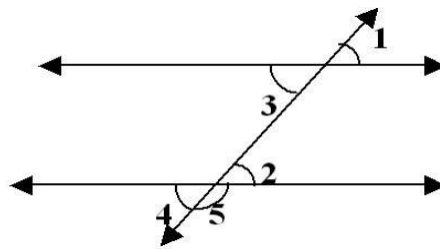
Ans: (c) $112^\circ, 68^\circ$

72. Find the angle, which is equal to its complement.

- (a) 30° (b) 25° (c) 35° (d) 45°

Ans: (d) 45°

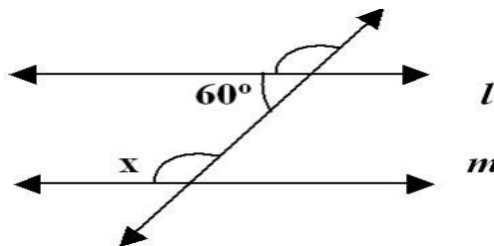
73. In fig. pair of alternate exterior angles are:



- (a) $\angle 1, \angle 3$ (b) $\angle 2, \angle 3$ (c) $\angle 2, \angle 5$ (d) $\angle 1, \angle 4$

Ans: (d) $\angle 1, \angle 4$

74. Lines $l \parallel m$, t is a transversal then $\angle x = ?$



- (a) 120° (b) 60° (c) 180° (d) 90°

Ans: (a) 120°

75. Find the angle, which is equal to its supplement.

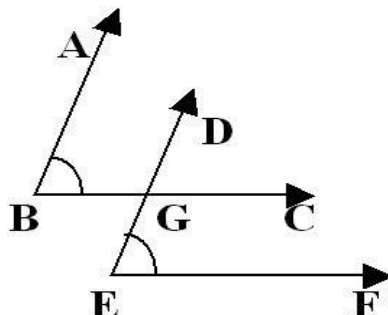
- (a) 60° (b) 90° (c) 180° (d) None of these

Ans: (b) 90°

76. The angle which is four times its complement is
 (a) 60° (b) 30° (c) 45° (d) 72°
 Ans: (d) 72°

$$\text{The equation } x = 4(90^\circ - x) \Rightarrow x = 360^\circ - 4x \Rightarrow x + 4x = 360^\circ \Rightarrow 5x = 360^\circ \Rightarrow x = 72^\circ$$

77. If arms of two angles are parallel, then find the $\angle DEF$

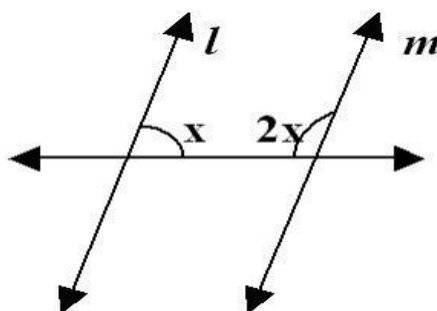


- (a) 15° (b) 90° (c) 180° (d) 75°
 Ans: (d) 75°

78. The angle which is five times its supplement is
 (a) 150° (b) 180° (c) 90° (d) 360°
 Ans: (a) 150°

$$x = 5(180^\circ - x) \Rightarrow x = 900^\circ - 5x \Rightarrow 6x = 900^\circ \Rightarrow x = 150^\circ$$

79. Find x if $l \parallel m$



- (a) 30° (b) 60° (c) 90° (d) 180°
 Ans: (b) 60°

80. Which pair of following angles are complementary?
 (a) $73^\circ, 27^\circ$ (b) $75^\circ, 25^\circ$ (c) $48^\circ, 52^\circ$ (d) $35^\circ, 55^\circ$
 Ans: (d) $35^\circ, 55^\circ$

81. Which pair of following angles are supplementary?
 (a) $110^\circ, 50^\circ$ (b) $105^\circ, 65^\circ$ (c) $50^\circ, 130^\circ$ (d) $45^\circ, 45^\circ$
 Ans: (c) $50^\circ, 130^\circ$

82. What is complement of 63° ?

- (a) 18° (b) 27° (c) 30° (d) 21°

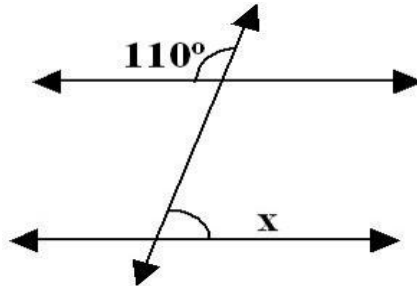
Ans: (b) 27°

83. Find the supplement of 105° .

- (a) 80° (b) 65° (c) 75° (d) 100°

Ans: (c) 75°

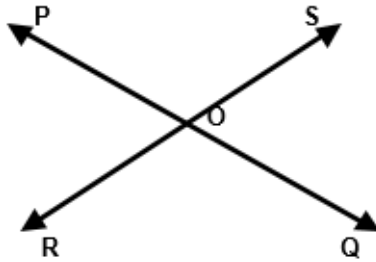
84. Find the value of x if $l \parallel m$



- (a) 110° (b) 70° (c) 90° (d) 180°

Ans: (b) 70°

85. Two lines PQ and RS intersect at O. If $\angle POR = 50^\circ$, then value of $\angle ROQ$ is



- (a) 120° (b) 130° (c) 90° (d) 150°

Ans: (b) 130°

86. The magnitude of the angle is measured in

- (a) Meter (b) Degree (c) Decibel (d) None

Ans: (b) Degree

87. The angle in a right angled triangle other than right angle is

- (a) Acute (b) Obtuse (c) Reflex (d) Straight

Ans: (a) Acute

88. When the sum of two angles is 90° then the angle are said to be

- (a) Straight (b) Complementary (c) Supplementary (d) Reflex

Ans: (b) Complementary

89. When the sum of two angles is 180° then the angle are said to be

- (a) Straight (b) Complementary (c) Supplementary (d) Reflex

Ans: (c) Supplementary

90. When the two angles lie on opposite side of a common arm and have a common vertex they are known as

- (a) Complementary (b) Supplementary (c) Adjacent (d) Interior

Ans: (c) Adjacent

91. Two adjacent supplementary angles form a ----- pair.

- (a) Adjacent (b) Line (c) Linear (d) Arm

Ans: (c) Linear

92. Vertically opposite angles are always...

- (a) Equal (b) Unequal (c) Can't say (d) None

Ans: (a) Equal

93. The complement of an angle 4° is

- (a) 85° (b) 90° (c) 86° (d) none

Ans: (c) 86°

94. When two lines intersect each pair of angle formed without common area is called Angle.

- (a) Adjacent (b) Vertically Opposite (c) Complementary (d) Supplementary

Ans: (b) Vertically Opposite

95. An angle is equal to its complement then the measure of angle is

- (a) 90° (b) 45° (c) 30° (d) 60°

Ans: (b) 45°

96. Sum of all angles around a point is 180°

- (a) True (b) False (c) Can't say (d) None

Ans: (b) False

97. The angle between two hands of a 6 'o clock is

- (a) 180° (b) 90° (c) 150° (d) 270°

Ans: (a) 180°

98. If the difference between two supplementary angles is 0, then one of the angle is

- (a) 103° (b) 110° (c) 90° (d) 115°

Ans: (c) 90°

99. The two supplementary angles are in the ratio of 5:13, and then the angles are

- (a) 50° , 130° (b) 110° , 70° (c) 60° , 120° (d) 25° , 155°

Ans: (a) 50° , 130°

100. When a transversal intersect a set of parallel lines, then....

- (1) Alternate angles are equal (2) Corresponding angles are equal.
(3) Sum of the two interior opposite angles on the same side of transversal is 360° .

(a) 1, 2, 3 are correct

(b) 2, 3 are correct

(c) 1, 2 are correct

(d) 1, 2, 3 are incorrect

Ans: (c) 1, 2 are correct